CHAPTER VII

4.3 (C) Can AI systems be moral agents? If so, how should we hold them accountable?

Can Artificial Intelligence (AI) systems function as moral agents? The inquiry pertains to their capacity for moral decision-making and accountability for their actions, akin to that of humans. To address this topic, we must examine both the essence of moral agency and the limits of AI autonomy. Moral agency generally requires the following conditions:
A moral agent can be defined as an entity capable of making moral judgements and being held ethically accountable for its acts. Moral agents possess the capacity to distinguish between good and wrong, act based on ethical principles, and are subject to commendation or censure for their conduct. The notion of moral agency is fundamental to ethical studies, as it elucidates responsibility, accountability, and the formulation of ethical decisions.
The possession of many essential capabilities is frequently necessary for an entity to be regarded as a moral agent. Included are consciousness, which allows awareness of one's actions and their consequences; intentionality, the capacity to formulate intentions and act for reasons; autonomy, the ability to exercise self-governance and make independent decisions; and moral reasoning, which enables the evaluation of actions in moral terms. Collectively, these capabilities enable an agent to understand the ethical significance of its acts.

Humans exemplify moral agents, as they can reflect on ethical principles, understand the repercussions of their acts on others, and assume accountability for their choices. Individuals might receive moral commendation or condemnation for their acts when they act with awareness and autonomy, which highlights the importance of moral responsibility in ethical discussions. Animals may possess moral standing owing to their sentience; yet, they are often not regarded as full moral agents due to their restricted capacity for moral reasoning, which limits their ability to make autonomous decisions and be held accountable for their actions in the same way that humans are.

Conversely, systems driven by artificial intelligence are often seen as amoral. While artificial intelligence can do complex tasks and follow set ethical guidelines, it lacks consciousness, true intentionality, and authentic moral understanding. The conduct of artificial intelligence systems is dictated by algorithms and human-defined objectives; these systems lack comprehension of moral principles and do not possess a sense of accountability for their actions.

Autonomy: The term "autonomy" refers to an individual's ability to make independent, informed, and voluntary judgements regarding their actions and life. In moral philosophy, autonomy is closely associated with notions such as self-governance, freedom of choice, and the capacity to act in alignment with one's own values and rationales, rather than being subjected to external coercion or control.

Autonomy is a basic principle in ethics, as it underpins human dignity and moral responsibility. An individual's autonomy must be honoured, encompassing recognition that they possess the liberty to make personal decisions, regardless of external dissent. In healthcare ethics, informed consent is essential, as it recognises and honours the patient's autonomy in decision-making. Autonomy is a crucial feature in the assignment of moral culpability. An individual can only be deemed morally accountable for their actions if they act autonomously, which is characterised by understanding, intention, and freedom. When actions are executed under coercion, manipulation, or profound ignorance, moral accountability is reduced or nullified. The term "autonomy" assumes a specific meaning within the realm of artificial intelligence. Although artificial intelligence systems can operate independently in a technical capacity, executing tasks without ongoing human intervention, they lack moral autonomy. Artificial intelligence systems lack consciousness, self-awareness, and the ability to engage in moral reasoning. Artificial intelligence systems do not base their decisions on self-chosen principles; instead, they rely on ingrained patterns and standards. Consequently, artificial intelligence lacks moral autonomy, even though it may possess operational autonomy. The conduct of artificial intelligence (AI) must remain under human oversight, encompassing designers, developers, and users who define objectives and constraints.

Intentionality: Intentionality: In the domains of philosophy of mind and ethics, the notion of intentionality—denoting the capacity of mental states to refer to or be directed toward an object—is regarded as one of the most essential ideas. Beliefs, wants, intentions, and aims exemplify intentional states, as they signify or denote entities, events, or consequences in the world. An illustration of this would be an individual intending to aid another, with their activities guided by an intrinsic goal and an understanding of their conduct. In the field of ethics, the notion of intentionality is crucial in delineating moral responsibility. The assessment of deliberate activities differs from the assessment of actions performed inadvertently or unconsciously. The outcome of an action influences the appropriateness of moral praise or condemnation, although the agent's intentions also contribute significantly; for instance, a person who intentionally harms another is typically judged more harshly than someone who

causes harm accidentally, despite similar outcomes. In terms of morality, wilfully inflicting harm on another individual is regarded as more unethical than causing harm inadvertently, even if the outcomes are the same. Consciousness and agency are closely linked to the notion of intentionality. The ability to formulate justifications for actions, possess consciousness, and attain understanding are essential elements of authentic intentionality. Humans possess intentionality due to their ability to reflect on their goals, evaluate the ethical rationales for their actions, and choose from multiple alternatives. Due to this power, they can be held morally accountable for their actions. Conversely, programmes utilising artificial intelligence lack authentic intentionality. Although artificial intelligence systems appear to be goal-orientated, their "intentions" are fundamentally rooted in algorithms, training data, and objectives established by humans. Artificial intelligence lacks understanding or endorsement of its goals; it simply executes the directives it has been trained to follow. Thus, artificial intelligence exhibits only as-if or derived intentionality, rather than authentic intentionality, because it does not possess consciousness or self-awareness to form genuine intentions or desires.

Reasoning and comprehension of morality: In the realm of ethics, moral reasoning or moral comprehension denotes the ability to recognise moral dilemmas, ascertain what is right and wrong, and justify actions by invoking ethical principles. The process involves understanding ethical concepts such as justice, fairness, rights, duties, and harm, and subsequently applying these principles to real-life situations.

Individuals can contemplate their actions and choose ethically appropriate behaviours when they engage in moral reasoning. Ethics and moral reasoning closely link the notions of consciousness, intentionality, and autonomy. A person proficient in moral thinking can understand the justification for actions, foresee the ethical consequences of decisions, and acknowledge the effects of those choices on others. Through social interaction, education, empathy, and cultural norms, individuals develop a moral understanding that allows them to feel guilt, sorrow, or pride regarding their actions. Moral reasoning is an essential element of moral responsibility within ethics. For an agent to receive praise or criticism, it is essential for them to understand the ethical implications of their acts. Actions such as deliberate deception or discrimination are morally reprehensible because the individual recognises that these behaviours violate ethical standards. Presently present artificial intelligence systems lack genuine moral reasoning or understanding, which means they cannot discern right from wrong or recognise the moral implications of actions such as deliberate deception or discrimination. This holds true in the realm of artificial intelligence. Artificial intelligence (AI) lacks a true

comprehension of moral notions; however, it can be programmed to adhere to ethical standards or enhance outcomes. Artificial intelligence systems lack consciousness and cannot grasp the ethical implications of actions; they merely evaluate data and execute algorithms. Consequently, their "ethical action" is a mere simulation rather than an authentic exhibition of reason. Capacity to understand theoretical concepts of morality, discern between right and wrong, and contemplate ethical issues.

Consciousness/Sentience: Sentience is defined as the capacity to have subjective experiences, including the ability to perceive pain, pleasure, emotions, or feelings. Consciousness is the condition of being cognisant of oneself and one's environment. Sentience and awareness are the two elements that collectively form the basis of the first-person interior experience commonly known as the experience of being an entity. These capacities are vital to their research since they establish a moral position. Conscious and sentient beings can endure misery and feel well-being, thereby providing moral justification to regard their interests. Humans and several animals are deemed ethically significant due to their capacity to experience both pain and pleasure, which underscores the moral obligation to consider their welfare in ethical decision-making.

A crucial element of moral responsibility is the existence of awareness and sensitivity. An agent can only be held ethically liable if it possesses awareness of its actions and the capacity to understand their significance. Without consciousness, actions become mechanical rather than purposeful, and professions of moral commendation or condemnation lose their relevance, as they require an understanding of ethical implications and the ability to make choices based on moral reasoning.

Current artificial intelligence systems are often considered to be non-conscious and non-sentient. Although artificial intelligence can analyse information, recognise patterns, and produce responses akin to human behaviour, it lacks subjective experience and emotions. Artificial intelligence lacks the capacity to experience emotions, including joy, pain, remorse, or empathy. Consequently, artificial intelligence is incapable of moral injury and lacks moral rights.

The question of whether artificial general intelligence could possess consciousness or sentience remains a topic of philosophical debate. The emergence of genuine awareness or sensitivity in artificial intelligence in the future would raise substantial ethical issues regarding the moral status, rights, and responsibilities of the technology. Nonetheless, this remains a subject of

speculation at present. Humans exhibit these traits due to our possession of consciousness, emotions, and moral reasoning capabilities.

Current artificial intelligence systems do not qualify as moral agents. Contemporary AI systems, regardless of their intelligence, are devoid of awareness and ethical comprehension. They function through algorithms and data patterns, rather than authentic moral contemplation. AI may replicate moral thinking (e.g., selecting the "most ethical" alternative in a problem), although it lacks comprehension of the moral significance of its decisions. Therefore, it cannot be considered a genuine moral agent in the human context.

Lack of Consciousness and Intentionality: Consciousness and intentionality are fundamental to moral philosophy and essential to establishing moral agency and accountability. Consciousness denotes an entity's self-consciousness and awareness of its environment, along with the ability for subjective experiences, including feelings, sensations, and emotions. Intentionality denotes the capacity to formulate objectives and possess mental states orientated towards goals, objects, or consequences.

Contemporary artificial intelligence (AI) systems are devoid of consciousness and authentic intentionality. While AI can analyse vast datasets, identify patterns, and execute judgements, these functions transpire devoid of consciousness or subjective experience. AI lacks the capacity to experience pain, pleasure, remorse, or empathy, and it does not comprehend the meaning or ethical implications of its actions. Its operations are exclusively computational.

Likewise, AI does not possess genuine intentionality. Although AI systems seem to be purpose-driven, their objectives are externally determined by human creators. AI lacks the capacity to select its objectives or comprehend the rationale for their pursuit; it simply adheres to programmed directives or optimises mathematical functions. This behaviour is frequently characterised as "as-if intentionality" rather than "authentic intentionality."

Due to AI's absence of consciousness and intentionality, it cannot be deemed a moral agent. Moral accountability necessitates cognisance of one's acts and the motives underlying them. In the absence of these capacities, AI cannot be commended, criticised, or held morally responsible for its outcomes. Consequently, any damage inflicted by AI systems—such as bias, discrimination, or abuses of rights—must be ascribed to the human agents responsible for their design, implementation, or oversight.

The lack of awareness and intentionality in AI elucidates why AI systems are more accurately regarded as instruments rather than moral agents. Ethical and legal accountability for AI conduct must stay unequivocally with people and institutions until, and unless, future iterations of AI acquire these foundational attributes.

Contemporary artificial intelligence systems, particularly the advanced ones, are generally regarded as devoid of genuine consciousness, subjective experience, or the capacity to formulate intentions like those of humans. Programming directives, algorithms, and training data are the elements that govern their actions.

Functional Imitation versus Genuine Comprehension: Although artificial intelligence can mimic moral conduct by generating ethical responses or avoiding pre-programmed harmful actions, this is regarded as functional equivalency rather than genuine moral awareness or agency. They operate not on ethical reasoning but on a set code, which means their responses are predetermined and lack the ability to understand the moral implications of their actions.

Contemporary artificial intelligence (AI) systems are devoid of consciousness and authentic intentionality. While AI can process extensive data, identify patterns, and make judgements, these functions occur without consciousness or subjective experience. AI lacks the capacity to experience pain, pleasure, remorse, or empathy, and it does not comprehend the meaning or moral implications of its actions. Its operations are exclusively computational.

Likewise, AI lacks genuine intentionality. Although AI systems seem to be purpose-driven, their objectives are externally determined by human creators. Artificial intelligence does not select its aims or comprehend its rationale for pursuing them; it simply adheres to programmed directives or optimises mathematical functions. This behaviour is frequently characterised as "as-if intentionality" rather than "authentic intentionality."

Due to AI's absence of consciousness and intentionality, it cannot be regarded as a moral agent. Moral accountability necessitates cognisance of one’s acts and the motivations underlying them. In the absence of these capacities, AI cannot be lauded, criticised, or held morally responsible for its outcomes. Consequently, any damage inflicted by AI systems—such as bias, discrimination, or abuses of rights—must be ascribed to the human agents responsible for their design, implementation, or oversight.

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Functional Imitation vs. Genuine Comprehension: Although artificial intelligence can replicate moral behaviour by generating ethically sounding responses or evading pre-programmed harmful actions, this constitutes functional equivalence rather than authentic moral consciousness or agency. They operate not on moral reasoning but on a codified system, which means they follow predefined rules and algorithms rather than engaging in true ethical deliberation or understanding.

Lack of Free Will: Free will is the ability of an agent to make choices freely among several options, to act voluntarily, and to exert control over one's decisions. The concept of free will has been thoroughly examined and articulated. A robust correlation exists between free will and moral responsibility within moral theory. An actor can only be deemed morally guilty if they have the capacity to act differently.

Currently, artificial intelligence systems lack free will. Although artificial intelligence systems can operate autonomously in a technical context, their activities are wholly governed by algorithms, training data, model design, and objectives set by people. Artificial intelligence does not do actions based on values it has chosen for itself or its objectives. Rather, it conforms to established standards or procedures that enhance performance.

Artificial intelligence, lacking free will, cannot make truly voluntary decisions. Artificial intelligence systems lack moral deliberation, disregard reasons as matters of value, and do not alter their actions based on moral reasoning. Any decision seemingly made by artificial intelligence is, in fact, the result of computational processes rather than an exercise of free will.

Due to the lack of free will, artificial intelligence cannot be considered a moral agent and cannot be held ethically accountable for its actions. One must possess the capacity for alternative choices and conscious behaviour to assign moral blame or praise. When artificial intelligence systems cause harm, including biased decisions or rights violations, accountability rests with the human designers, developers, deployers, and regulators who influenced the system's behaviour.

A self-governing will, often a prerequisite for moral responsibility, is absent in current artificial intelligence; instead, AI acts are ultimately determined by their design and data.

It is important to note that AI, although termed a "moral actor", does not possess moral agency. Although AI lacks moral agency, it can still engage in actions that yield moral consequences. Moral problems arise in scenarios when an autonomous vehicle determines whom to save during an accident or when an algorithm assesses loan eligibility. Consequently, artificial intelligence can be regarded as a moral actor due to its influence on ethical scenarios; nevertheless, it cannot be classified as a moral agent since it bears no responsibility for those occurrences.

Philosophers may contemplate granting moral agency to artificial intelligence systems if, in the future, these systems achieve authentic self-awareness or moral reasoning. In this context, they may be held accountable through digital rights, legal personhood, or established ethical norms; yet, this remains a disputed and theoretical issue, as the criteria for determining self-awareness and moral reasoning in AI systems are still under debate among ethicists and legal scholars.

Accountability is predominantly assigned to humans during the AI lifecycle, as the current generation of artificial intelligence lacks full moral agency. It is crucial to underscore the principle of divided responsibility. Accountability is typically achieved through a multilayered framework that prioritises governance, transparency, and regulation.

Distributed Responsibility

Accountability is a collective responsibility among the diverse human stakeholders. In the realm of artificial intelligence, accountability is distributed among human stakeholders because AI systems lack consciousness, intentionality, and free will. Accountability for artificial intelligence operations must rest with the individuals engaged in its creation, deployment, and oversight. This is because AI cannot act as a moral agent. Consequently, accountability is not assigned to the computer but is instead distributed among the individuals.

Designers and developers

Designers and developers are responsible for the building of artificial intelligence systems. These behaviours include the selection of training data, the formulation of objectives, the integration of values, and the foresight of potential damages, such as prejudice or bias. The

accountability for an artificial intelligence system that exhibits bias or inflicts foreseeable harm partially resides with the individuals who developed and trained the system. Designers and developers bear significant moral and legal responsibilities for the ethical operation of artificial intelligence systems. Designers and developers play a crucial role in ensuring the ethical operation of these systems. Artificial intelligence systems cannot be held accountable for their actions, as they lack consciousness, intentionality, and free will. Consequently, it is the obligation of the individuals tasked with planning, constructing, and training these systems.

Designers and developers are tasked with delineating the objectives and constraints of artificial intelligence systems. They participate in the process of creating the model architecture, choosing the training data, and establishing the optimisation objectives. These decisions directly influence the conduct of an artificial intelligence system and the types of outcomes it produces. An illustration of this is that biased or unrepresentative training data might yield discriminatory outputs, so rendering designers ethically accountable for predicted harms.

Moreover, it is their obligation to integrate ethical principles into artificial intelligence systems using approaches such as fairness-aware design, transparency, explainability, and safety testing. It is essential for developers to do crucial tasks, including bias audits, risk assessments, and ongoing monitoring, to avert ethical harm and violations of rights.

Furthermore, it is incumbent upon designers and developers to anticipate misuse and mitigate applications that may be harmful. This includes the establishment of safeguards, the recording of system limitations, and the clear communication of risks to users and organisations implementing artificial intelligence. If this is not accomplished, it may result in a deployment that is either unethical or hazardous.

Organisations and deployers

Organisations and deployers are essential to ensuring the ethical use of artificial intelligence. This arises from their authority in establishing the techniques, locations, and objectives for the deployment of AI systems, which directly influences how these technologies are integrated into various sectors and ensures they align with ethical standards. They are responsible for implementing artificial intelligence technology in real-world settings. They ascertain the applicability, timing, and objectives of artificial intelligence deployment. The lack of adequate human oversight, supervision, or safeguards may result in misuse or harm, making these individuals morally and legally responsible. Despite a meticulously crafted artificial intelligence system, unethical or careless implementation might result in considerable moral detriment, making firms personally liable for AI consequences.

Organisations are responsible for identifying relevant use cases for AI. Implementing artificial intelligence in crucial sectors such as healthcare, law enforcement, recruitment, or credit assessment necessitates extreme prudence, openness, and human oversight. The improper application of artificial intelligence, whether by surpassing its designated parameters or neglecting its constraints, may result in discrimination, infringement of rights, or hazardous decision-making.

Moreover, employers must guarantee human-in-the-loop supervision. Fully automating critical judgements is unsuitable when such determinations could endanger human rights, dignity, or welfare. Organisations must provide procedures that enable individuals to review, reverse, or challenge decisions made by artificial intelligence.

Organisations are also accountable for monitoring, auditing, and updating AI systems after implementation. Real-world scenarios are ever-changing, and models may deteriorate or react unpredictably over time. Ethical negligence transpires when an individual disregards acknowledged performance discrepancies or hazards.

A significant obligation for AI accountability resides with organisations and implementers. Since AI systems lack moral agency, accountability for their real-world consequences lies with the organisations that choose to install and rely on them. The ethical implementation of artificial intelligence necessitates responsible governance, ongoing assessment, and compliance with human values.

Policymakers and regulators

The burden for ensuring the ethical, legal, and socially acceptable use of artificial intelligence rests firmly on legislators and regulators. Furthermore, they play a significant role. They are responsible for establishing ethical standards, legislative frameworks, and enforcement mechanisms to ensure that artificial intelligence systems respect public values, safety, and rights. Insufficient or absent regulation may allow detrimental AI methodologies to persist. Policymakers and regulators are tasked with establishing external frameworks that delineate accountability and protect public interests. This is because artificial intelligence systems are not moral agents and cannot be held responsible for their actions.

The primary objective is to establish well-defined regulations, standards, and ethical principles governing the research and implementation of artificial intelligence. This area encompasses regulations related to data protection, privacy, fairness, transparency, safety, and non-discrimination. Without adequate oversight, there exists a risk that harmful or immoral AI actions may remain undetected.

Enforcement and oversight are the responsibilities of both policymakers and regulators. Accountability systems can be established to assure that businesses and developers adhere to ethical and legal standards. These processes may encompass audits, impact assessments, reporting obligations, and potential punishments. Robust enforcement facilitates the prevention of moral harm and rights abuses.

Simultaneously, authorities are charged with balancing the preservation of fundamental rights and fostering innovation. Alongside promoting technological progress, they must ensure that artificial intelligence systems uphold human dignity, autonomy, and equality. This requires the continual updating of policies in response to advancements in AI technologies.

Policymakers and regulators serve as custodians of public values inside the AI ecosystem. By establishing regulations, enforcing accountability, and protecting human rights, they guarantee that the responsibility for AI systems resides with human institutions rather than being transferred to computers.

Users and operators

Users and operators are essential stakeholders in the ethical use of artificial intelligence, as they engage directly with AI systems and depend on their outputs for real-world decision-making. Moreover, the way users and operators comprehend and execute AI recommendations can significantly influence the results, even though AI systems lack moral agency.

Users and operators are responsible for utilising artificial intelligence systems in a responsible and appropriate manner. A thorough understanding of the system's objectives, limitations, and any biases is necessary for this. Blindly trusting or utilising artificial intelligence's (AI) outputs beyond their intended parameters can result in ethical harm, including activities that infringe on rights or yield unjust outcomes.

They bear the obligation to apply human judgement and oversight, especially in critical scenarios such as those found in education, law enforcement, and healthcare. AI outputs should enhance human decision-making, not replace it. Individuals possessing the authority to contest, supersede, or escalate decisions rendered by artificial intelligence are required to act if ethical dilemmas emerge, such as situations where AI may produce biased outcomes or make decisions that conflict with human values.

Moreover, users and operators have equal responsibility for reporting any errors, misuse, or adverse outcomes generated by the system. Feedback from real-world applications is essential for enhancing artificial intelligence systems and mitigating repeated harm, as it allows developers to identify and address specific shortcomings and improve user safety and system

reliability. Persisting in the use of a system recognised as having deficiencies or disregarding established issues constitutes ethical negligence.

Both users and operators share the duty of using artificial intelligence judiciously, making informed decisions, and diligently overseeing its performance. The ethical obligation for artificial intelligence systems includes the individuals who utilise and depend on them in daily operations. This is due to the fact that AI systems cannot be deemed morally accountable.

Knock your brain

1. A moral agent is best defined as an entity that can:

a) Perform complex calculations

b) Follow programmed rules

c) Distinguish right from wrong and be held responsible

d) Store information

2Moral agents must possess:

a) Automation and efficiency

b) Intentionality and autonomy

c) Network connectivity

d) Hardware reliability

3 Which ethical capacity allows a moral agent to be praised or blamed?

a) Storage capacity

b) Free will

c) Automation

d) Pattern recognition

4When AI causes harm, who should be held morally accountable?

a) The AI system

b) The algorithm

c) Human stakeholders involved

d) Random chance

5.If an AI system were conscious and morally autonomous, it might:

a) Still be a tool

b) Be considered a moral agent

c) Lose efficiency

d) Stop learning

Unleash your potential

1.Assess the extent to which machines can genuinely exhibit moral agency.

2.Examine the ethical dilemmas associated with attributing moral agency to computers.